

Retail Business Performance Analysis

A business intelligence assessment of vendor, brand, purchasing, sales, and inventory performance

Purchase Quantity	32M	Total Purchases	\$307.34M
Sales Quantity	32M	Total Sales	\$441.41M
Stock Turnover	17.40K	Unsold Capital	\$2.71M
Profit Margin	331.62K	Gross Profit	\$134.07M

Executive Summary

This report brings together purchasing, sales, product pricing, freight, and inventory information to reveal what drives revenue, margin, and working-capital performance. SQL creates a reusable summary table, Python performs analysis and statistical checks, and Power BI turns the findings into a decision-ready dashboard.

Scope

The project demonstrates an end-to-end analytics workflow: data modeling, derived business measures, exploratory analysis, statistical comparison, and an executive dashboard.

Tools: SQLite, Python, Pandas, Matplotlib, Seaborn, SciPy, and Power BI

Business Problem

Retail data is often fragmented across purchase transactions, sales records, price lists, vendor invoices, and inventory snapshots. This makes it difficult to decide where to invest, which suppliers to prioritize, and where capital is being tied up.

The project creates a vendor-and-brand performance layer so leaders can compare commercial value, procurement behavior, and inventory efficiency in one place.

Business questions

- Which vendors and brands generate the most sales revenue?
- Which brands have low sales but high margins and should be tested for promotion?
- How concentrated is procurement spending among leading vendors?
- Does ordering in larger quantities reduce average unit cost?
- Which vendors have the greatest unsold inventory and tied-up capital?
- Do high- and low-performing vendors have different profit-margin patterns?

Business impact

The analysis supports vendor negotiations, promotion planning, assortment decisions, reorder control, and a more resilient supplier strategy.

EDA Insights

The EDA establishes data quality and builds an analytical table before comparing performance segments.

Data preparation

- SQL common table expressions aggregate purchase, sales, and freight data and join them into Business_Performance_summary.
- Vendor and description fields are trimmed, volume is converted to a numeric data type, and missing values are handled for analysis.
- Records with non-positive gross profit, profit margin, or sales quantity are excluded from the performance analysis.

Measures created

Metric	Calculation
Gross Profit	Total Sales Dollars - Total Purchase Dollars
Profit Margin	Gross Profit / Total Sales Dollars x 100
Stock Turnover	Total Sales Quantity / Total Purchase Quantity
Sales-to-Purchase Ratio	Total Sales Dollars / Total Purchase Dollars
Unsold Inventory	Total Purchase Quantity - Total Sales Quantity

What the EDA revealed

- Business metrics are skewed with meaningful outliers, especially in freight and product-price measures.
- Purchase and sales quantities have a very strong positive relationship (approximately 0.999), confirming the importance of sell-through planning.
- Profit margin and sales price have a weak negative relationship (approximately -0.179), so price increases alone are not a margin strategy.

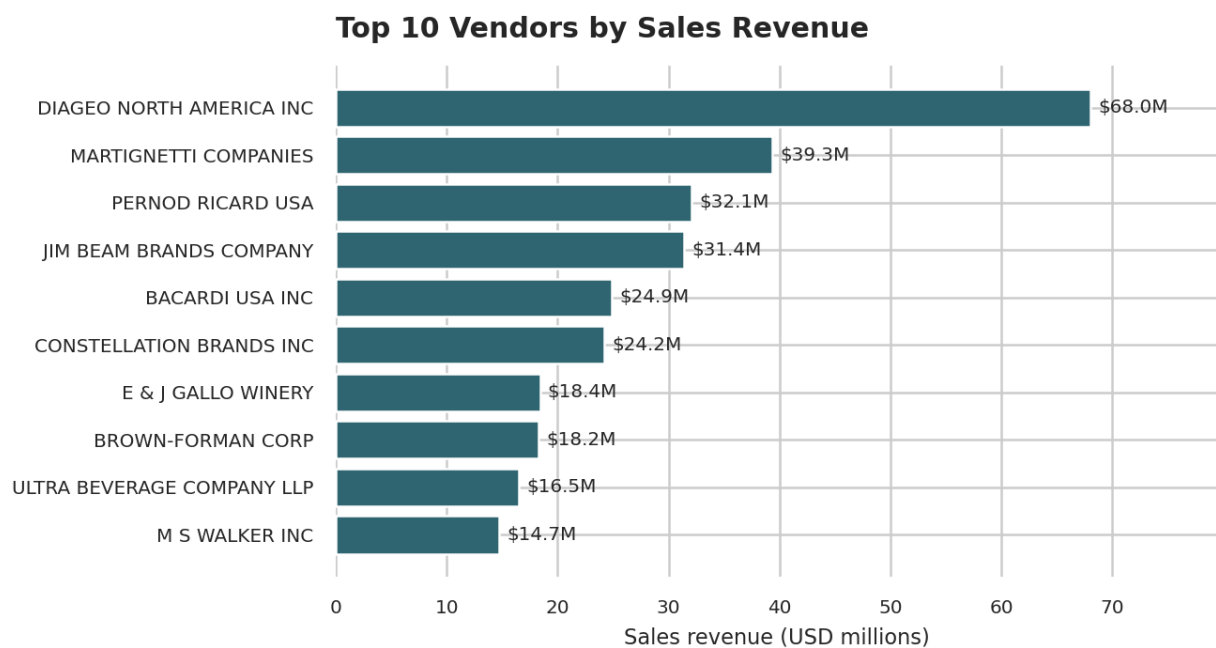
Summary Statistics

	Count	Mean	Median	Minimum	Maximum
Purchase Qty	8,564.00	3,772.01	431.00	1.00	337,660.00
Purchase \$	8,564.00	35,887.25	5,018.40	0.71	3,811,251.60
Sales Qty	8,564.00	3,748.98	490.50	1.00	334,939.00
Sales \$	8,564.00	51,542.28	8,483.78	1.98	5,101,919.51
Gross Profit	8,564.00	15,655.02	3,087.05	0.04	1,290,667.91
Profit Margin	8,564.00	38.72	33.36	0.02	99.72
Stock Turnover	8,564.00	2.03	1.00	0.43	274.50

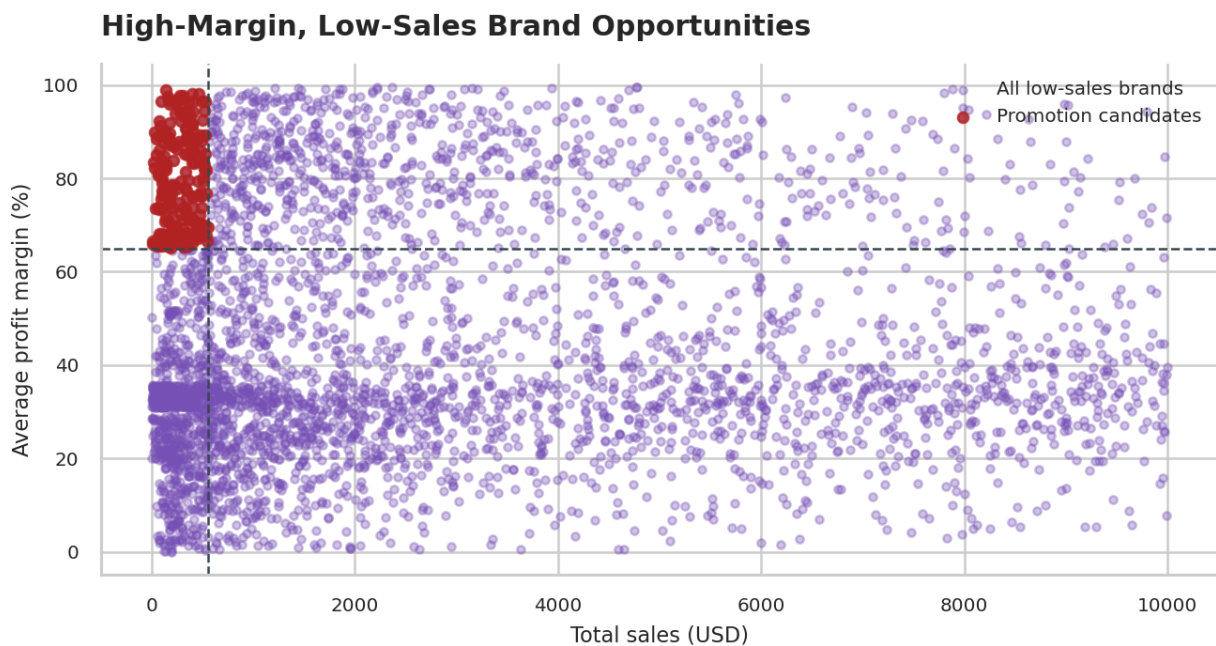
The cleaned dataset contains 10,692 vendor-brand observations. The table highlights the spread in purchase, sales, profit, and turnover measures, supporting the outlier checks completed during EDA.

Built Analytical Visuals

Included visuals were generated from the consolidated business-performance table using the same analytical logic as the notebook.



Top vendors produce a large share of sales. This is commercially valuable but should be paired with supplier-dependency monitoring.



Red points show high-margin, low-sales brands - a targeted opportunity segment for promotion, placement, and pricing experiments.

Research Questions & Key Findings

The analysis combines descriptive exploration with confidence-interval comparison of profit margins for top- and low-performing vendors.

Which vendors lead sales?

Leading vendors contribute a substantial share of revenue. Diageo North America, Martignetti Companies, Pernod Ricard USA, Jim Beam Brands, and Bacardi USA appear among the principal sales contributors.

Where is procurement concentrated?

The top 10 vendors account for about 65.69% of total purchases. This concentration should be monitored to reduce supplier-dependency risk.

Does bulk buying lower unit cost?

Larger purchase quantities generally show lower unit costs. The pattern supports negotiated bulk buying, but only where demand, storage, shelf life, and cash flow justify it.

Which brands are promotion candidates?

The high-margin/low-sales screen identifies 198 brands with untapped potential. These brands are suitable for focused campaigns rather than broad, untested promotions.

Where is capital tied up?

The dashboard reports \$2.71M in unsold capital. High-inventory vendors warrant review of assortment, reorder rules, clearance plans, and sales support.

How do performance segments differ?

Low-performing vendors have a 95% profit-margin confidence interval of 40.48%-42.62%, versus 30.74%-31.61% for top-performing vendors. This points to different volume, pricing, or cost structures.

Final Recommendations

Recommendations below convert the findings into an operational action plan.

Protect strategic revenue

Maintain strong relationships with leading sales vendors while tracking service, price, and margin contribution.

Reduce supplier concentration risk

Set procurement concentration thresholds and qualify secondary suppliers for critical categories.

Target high-margin, low-sales brands

Use focused promotions, bundles, better placement, and controlled pricing tests before expanding the assortment.

Release working capital

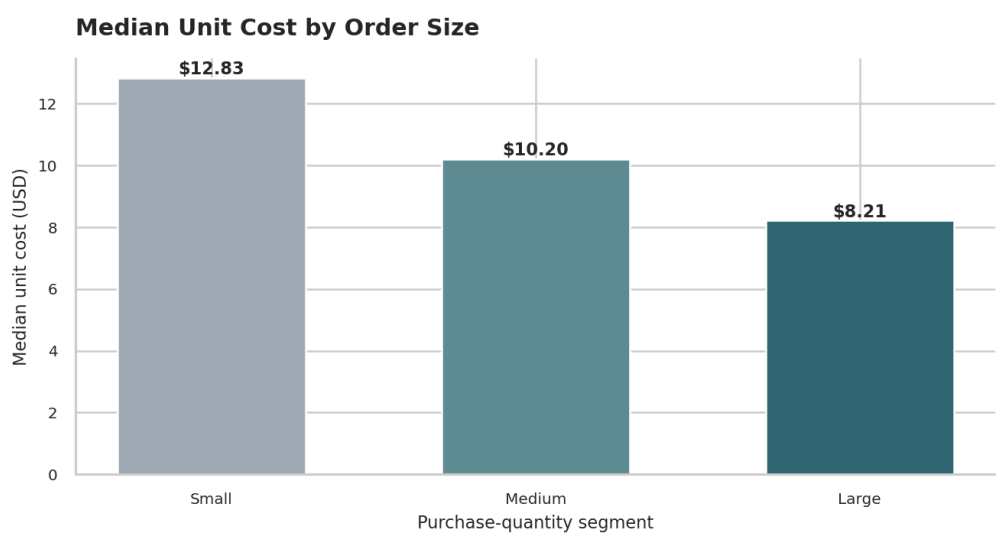
Prioritize slow-moving brands and vendors for clearance, reorder reduction, or assortment rationalization.

Apply bulk buying selectively

Use unit-cost savings together with demand forecasts, shelf life, storage capacity, and cash-flow impact.

Make reviews routine

Refresh the Power BI dashboard and use vendor scorecards that bring together sales, gross profit, margin, stock turnover, freight, and unsold inventory.

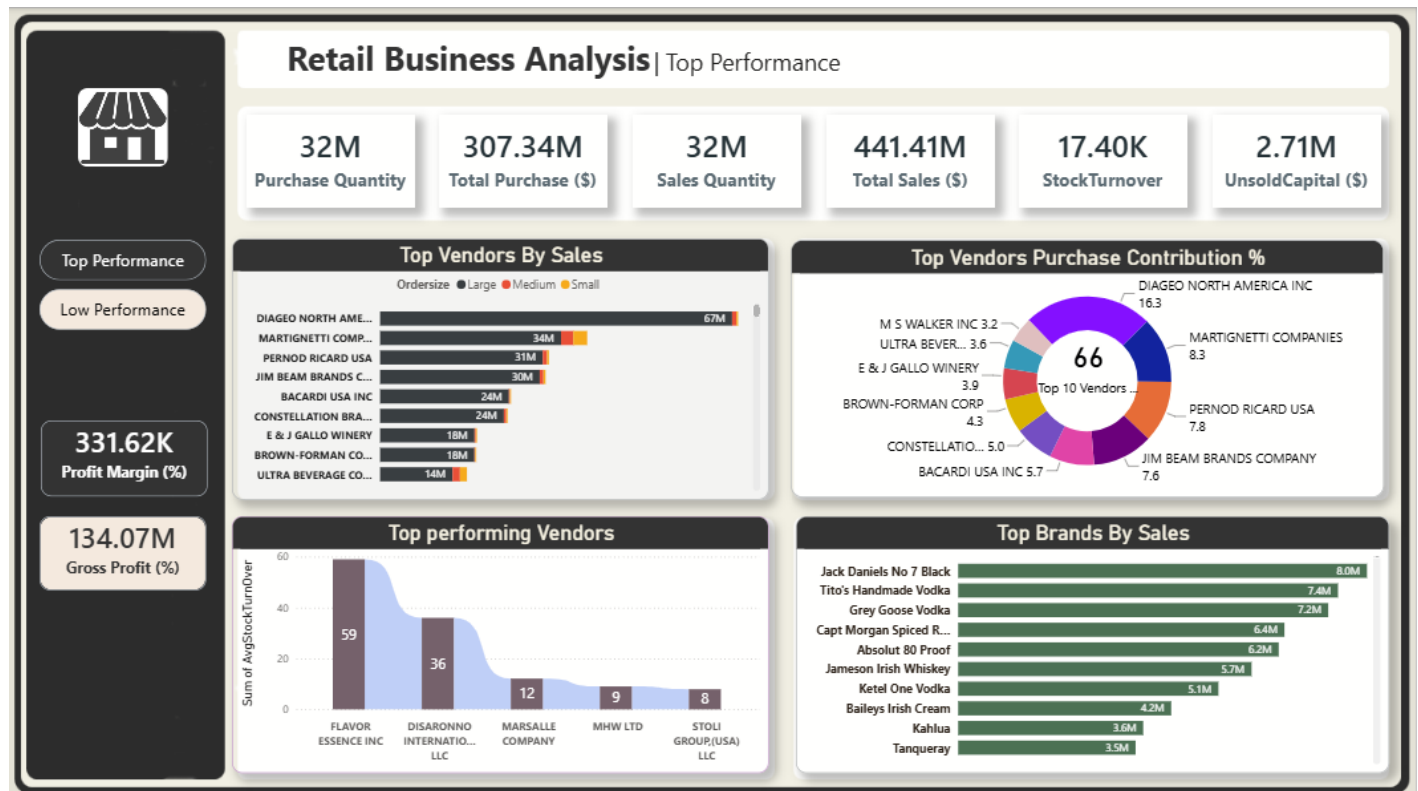


Recommended next step: establish a monthly vendor-performance review to identify exceptions and assign corrective actions.

Dashboard Appendix

Power BI provides complementary views of top and low performance. Both dashboard views are included for final reference.

Top Performance View



Low Performance View

